

Java Notes

03 January 2025 00:33

```
class Simple{
    public static void main(String args[]){
        System.out.println("Hello Java");
    }
}
```

Example to understand the types of variables in java

```
public class A
{
    static int m=100;//static variable
    void method()
    {
        int n=90;//local variable
    }
    public static void main(String args[])
    {
        int data=50;//instance variable
    }
} //end of class
```

Java Ternary Operator

In Java, the ternary operator is a concise way to write conditional expressions. It's often used as a shorthand for simple if-else statements.

java

 Copy code

```
condition ? expression1 : expression2;
```

- **condition:** A boolean expression (something that evaluates to true or false).
- **expression1:** The value returned if the condition is true.
- **expression2:** The value returned if the condition is false.

```
public class OperatorExample{
    public static void main(String args[]){
        int a=2;
        int b=5;
        int min=(a<b)?a:b;
        System.out.println(min);
    }
}
```

Output:

Swich Operation

Java for Loop

For loops in Java are a fundamental control structure used to repeat a block of code a specific number of times or iterate through a sequence of values. They are incredibly useful for tasks that require repetition, such as processing items in an array, generating repetitive output, or executing a block of code a predetermined number of times.

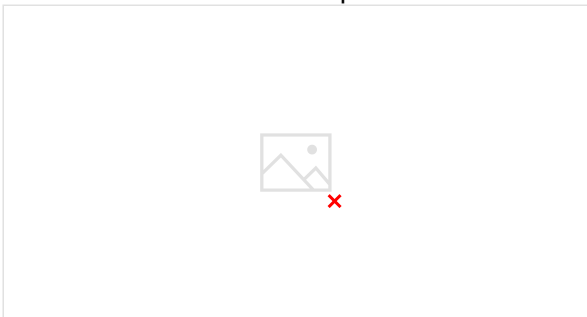
The Java *for loop* is used to iterate a part of the program several times. If the number of iteration is **fixed**, it is recommended to use for loop.

There are the following three types of for loops in Java.

- Simple for Loop
- [For-each](#) or Enhanced for Loop
- Labelled for Loop

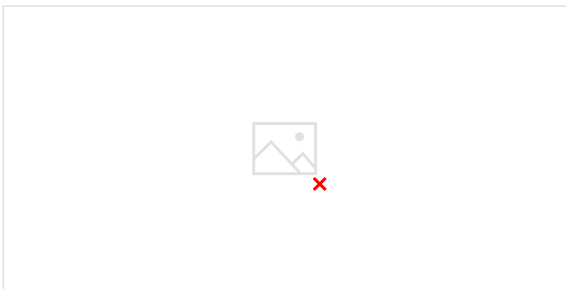


Java Nested for Loop



Java While Loop

The Java while loop is used to iterate a part of the program repeatedly until the specified Boolean condition is true. As soon as the Boolean condition becomes false, the loop automatically stops.





Deep Copy of Arr is generated in arr2 using clone() function





×



×





x



x



x



x



x



x



x



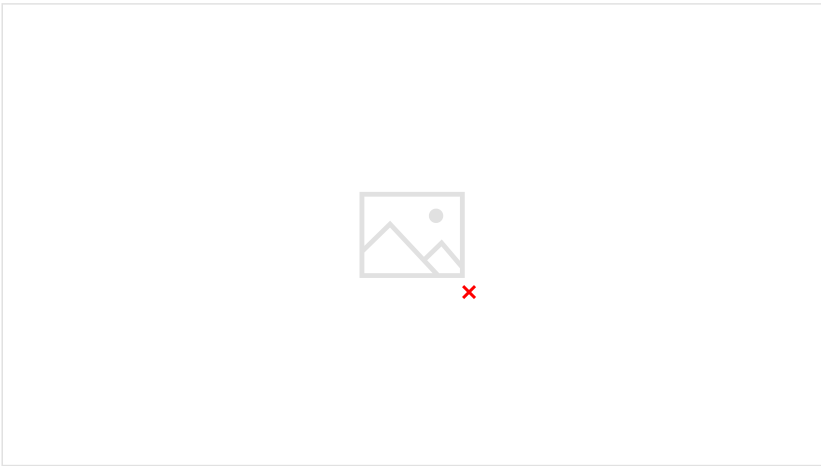
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×



×

Two Pointer Approach



×

Prefix Sum Approach



